ESE 4125 Question Map

Every question from the 2020, 2021, 2022, 2023, 2024 term papers, grouped by topic and ranked by how often it comes back and how many marks it carries. Answers are in the ESE 4125 Study Pack (HTML and PDF) next to this file.

24-hour list | Pass plan | Frequency grid | Section A | Section B

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 6 h for Part A and 5 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2024 ×5, 2023 ×4, 2022 ×3, 2021 ×2, 2020 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Next if you have time: the three short numericals (A3 dead time, A15 AFR, B5 TWA) take 10 minutes each with the calculators and give sure marks. This course is almost all lists: read each memory trick box twice, then write the list from memory once.

Section A: Occupational safety, hazards, fire, risk, PPE, accidents and EPP

Block 1: very likely and easy · 1.7 h

Topic	Chance	Level	Covered
A14 Causes of accidents: Heinrich's domino theory, 88-10-2 rule, man-man, man-machine and man-environment relationships	Very likely 100%	Easy	8%
A1 Basic definitions: hazard, risk, accident, occupational safety, health, safety management; the three parts of OSHEM	Very likely 100%	Easy	15%

A15 Accident investigation and reporting sheet; accident statistics AFR, ASR, FAR (with numerical); HAZAN, HAZOP; safety audit	Very likely 93%	Easy	25%
A2 Sources of hazard, categories of hazard by working situation, consequences, control steps, psychosocial hazards	Very likely 93%	Easy	34%

Block 2: very likely, medium effort · 0.8 h

Topic	Chance	Level	Covered
A10 Hazard identification methods: checklist, what-if, PHA, HAZOP, FMEA, plant drawings; limitations of hazard evaluation	Very likely 87%	Medium	42%

Block 4: likely · 3.5 h

Topic	Chance	Level	Covered
A13 Personal protective equipment (PPE): types, how each protects, respiratory protective equipment	Likely 73%	Easy	48%
A7 Classes of fire and fire-fighting materials (water, foam, dry powder, CO_2, wet chemical); fire-fighting procedure	Likely 67%	Easy	55%
A16 Emergency preparedness plan (EPP), Triple Bottom Line (TBL), safety committee	Likely 67%	Easy	61%
A11 Anatomy of a process incident; event tree vs fault tree	Likely 67%	Medium	65%
A8 BLEVE (boiling liquid expanding vapour explosion): mechanism, sequence of events, effects, prevention	Likely 67%	Medium	69%
A12 Risk management, hierarchy of controls (and where PPE fits), bow-tie diagram	Likely 60%	Medium	76%

✔ Pass line: 76% of Section A marks covered, enough to choose 3 of the 4 questions (about 5.9 h)

Only if time is left · 2.4 h

Topic	Chance	Level	Covered
A6 Fire basics: fire triangle and tetrahedron, factors contributing to fire, methods of extinguishment, types of explosion	Possible 53%	Easy	81%
A4 Physical and thermal hazards: heat cramps, heat stroke, frostbite, asphyxiation, cave in, radiation, lighting, housekeeping	Possible 47%	Easy	90%
A5 Electrical hazards (severity factors, protection) and biological hazards	Possible 47%	Easy	94%
A9 Fire management in a high-rise building; PASS	Possible 47%	Easy	97%
A3 Numericals: hazard matrix rating, fractional dead time and hazard rate of a protective system	Possible 33%	Medium	100%

Section B: Occupational health, chemicals, pollution and environmental management

Block 1: very likely and easy · 4.2 h

Topic	Chance	Level	Covered
B8 MSDS (material safety data sheet): purpose, 16 headings, key sections, responsibilities of supplier, employer, worker	Very likely 100%	Easy	11%
B11 Greenhouse effect and global warming, acid rain, smog: causes, effects, control	Very likely 100%	Easy	22%
B6 Hazardous chemicals: categories and health effects; hazardous vs toxic; routes of entry; local/systemic and acute/chronic effects	Very likely 100%	Easy	31%
B1 Occupational health, OSH Act and OSHA, occupational health service (functions, how to provide)	Very likely 100%	Easy	40%
B2 Occupational diseases: definitions, vs normal disease; lung disease, blood disorders, urticaria, pigmentary changes, sore eye, stress	Very likely 100%	Easy	49%
B3 Occupational ergonomics: definition, importance, five examples of good ergonomics	Very likely 100%	Easy	54%
B12 Sustainable development: definition, environmental/economic/socio-political sustainability, pollution prevention and clean technology, challenges	Very likely 93%	Easy	60%
B4 Occupational exposure, TLV and PEL, TLV-TWA, TLV-STEL, TLV-C	Very likely 93%	Easy	65%
B15 ISO 14001: purpose and major components of an environmental management system	Very likely 93%	Easy	70%
B10 Local, regional and global environmental impacts and their causes (thermal power plants, petroleum industry)	Very likely 93%	Easy	74%

Block 2: very likely, medium effort · 1.5 h

Topic	Chance	Level	Covered
B13 IEE and EIA: why needed, differences, EIA process (flow chart), red category, EMP	Very likely 100%	Medium	84%
✔ Pass line: 84% of Section B marks covered, enough to choose 3 of the 4 questions (about 4.9 h)			
B14 Environmental clearance in Bangladesh (Green, Orange-A, Orange-B, Red), Department of Environment (DoE) and its history	Very likely 93%	Medium	90%

Block 4: likely · 0.4 h

Topic	Chance	Level	Covered
B9 Pollution and pollutants: classification, sources of air and water pollution and their control	Likely 67%	Easy	95%

Only if time is left · 0.8 h

Topic	Chance	Level	Covered
B7 Dose-response relationship; lethal, sub-lethal, acute	Possible 47%	Easy	98%
B5 Numerical: time-weighted average (TWA) exposure and comparison with TLV	Possible 33%	Easy	100%

Pass plan

2021-2024 papers: 4 questions of 35 marks per section, answer 3 per section (full marks 210). 2020 was a short paper (3 questions of 30 per section). Section A = occupational safety: definitions and OSHEM, hazards and their control, hazard matrix, physical/thermal/electrical/biological hazards, fire and explosion, BLEVE, high-rise fire, hazard identification (PHA, HAZOP), anatomy of an incident, risk management and bow-tie, PPE, accident causation, accident statistics, EPP, TBL and safety committee. Section B = occupational health and environment: occupational health and diseases, ergonomics, exposure limits (TLV, PEL, TWA), hazardous chemicals, dose-response, MSDS, pollution, environmental impacts, greenhouse effect, acid rain, smog, sustainable development, IEE/EIA/EMP, environmental clearance in Bangladesh and ISO 14001. Materials: the textbook "Management of Occupational Safety, Health and Environment", Yasir Arafat's slides (fire, hazard identification, accidents), the PPE slides, Hasan's slides (occupational health, toxicity, MSDS, pollution and EIA), the teacher's Bangla notes (EPP, TBL) and Mostafizur Rahaman's summary notes. Almost everything is theory; only three short numericals exist (FDT/hazard rate, AFR and TWA). Each answer starts with a plain-language "Before you start" box and ends with a memory trick.

SECTION A: OCCUPATIONAL SAFETY, HAZARDS, FIRE, RISK, PPE, ACCIDENTS AND EPP

SECTION B: OCCUPATIONAL HEALTH, CHEMICALS, POLLUTION AND ENVIRONMENTAL MANAGEMENT

13 topics

cover 90% of all Section A marks asked

- Definitions and OSHEM, hazard categories and control, hazard matrix + FDT numerical, physical/thermal/electrical/biological hazards
- Fire triangle, explosions, fire classes and extinguishers, BLEVE, high-rise fire management
- Hazard identification (PHA, HAZOP, drawings), anatomy of an incident, fault and event trees, risk management, hierarchy, bow-tie, PPE
- Heinrich and man-machine, accident investigation and reporting, AFR/ASR/FAR + numerical, safety audit, EPP, TBL, safety committee

14 topics

cover 98% of all Section B marks asked

- Occupational health, OSHA, diseases (lung, blood, skin, eye, stress), ergonomics
- Exposure, TLV/PEL/TWA/STEL/C + TWA numerical, hazardous chemicals, routes, effects, dose-response, MSDS
- Pollutants, local/regional/global impacts, greenhouse effect, acid rain, smog, sustainable development
- IEE, EIA process, EMP, environmental clearance and DoE, ISO 14001

- **This course is mostly theory that repeats word for word.** Learn the definitions (A1), hazard categories (A2), anatomy of an incident (A11), bow-tie and hierarchy of controls (A12), PPE (A13), Heinrich and man-machine (A14), AFR/ASR/FAR (A15) and EPP steps (A16): these cover most of Section A.
- **Fire block (A6-A9):** fire triangle, extinguishing methods, fire classes, which extinguisher on which fire, BLEVE and high-rise fire management. One diagram each.
- **Section B is lists:** lung/blood/skin diseases (B2), ergonomics examples (B3), TLV types (B4), routes and effects of chemicals (B6), MSDS sections and responsibilities (B8), acid rain and smog (B11), EIA process flow chart (B13), red-category clearance (B14) and ISO 14001 components (B15).
- **Three tiny numericals:** FDT = F T / 2 and H = D × FDT (A3), AFR = injuries × 10^6 / man-hours (A15), TWA = $\Sigma C\ t\ /\ 8$ (B5). Each is 4-5 lines.
- **Always draw:** fire triangle/tetrahedron, BLEVE tank, anatomy table, event tree and fault tree, bow-tie, hierarchy pyramid, domino row, dose-response curve, greenhouse effect, acid rain cycle, EIA flow chart, red-category clearance chart.

Frequency grid

Marks asked from each topic in each paper. Darker means more marks. Click a topic to jump to its questions.

Marks per paper: 1–8 9–16 17–24 25+

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION A: OCCUPATIONAL SAFETY, HAZARDS, FIRE, RISK, PPE, ACCIDENTS AND EPP:

Topic	2024	2023	2022	2021	2020	Years	Marks	
A14 Causes of accidents: Heinrich's domino theory, 88-10-2 rule, man-man, man-machine and man-environment relationships	10	7	13	10	12	5/5	52	Must st
A1 Basic definitions: hazard, risk, accident, occupational safety, health, safety management; the three parts of OSHEM	12	19	7	5	6	5/5	49	Must st
A15 Accident investigation and reporting sheet; accident statistics AFR, ASR, FAR (with numerical); HAZAN, HAZOP; safety audit	15	7	24	25		4/5	71	Must st
A2 Sources of hazard, categories of hazard by working situation, consequences, control steps, psychosocial hazards	12	6	35	10		4/5	63	Must st
A10 Hazard identification methods: checklist, what-if, PHA, HAZOP, FMEA, plant drawings; limitations of hazard evaluation	10	5	23		17	4/5	55	Must stu
A13 Personal protective equipment (PPE): types, how each protects, respiratory protective equipment	10		10	10	8	4/5	38	Must

A7 Classes of fire and fire-fighting materials (water, foam, dry powder, CO_2, wet chemical); fire-fighting procedure	13	13	10	14	4/5	50	Must
A16 Emergency preparedness plan (EPP), Triple Bottom Line (TBL), safety committee	10	21		5	3/5	36	Must
A11 Anatomy of a process incident; event tree vs fault tree	13		12	6	3/5	31	Must s
A8 BLEVE (boiling liquid expanding vapour explosion): mechanism, sequence of events, effects, prevention	10		10	8	3/5	28	Must s
A12 Risk management, hierarchy of controls (and where PPE fits), bow-tie diagram	19	30			2/5	49	Must s
A6 Fire basics: fire triangle and tetrahedron, factors contributing to fire, methods of extinguishment, types of explosion	11			7	16	3/5	34	Must s
A4 Physical and thermal hazards: heat cramps, heat stroke, frostbite, asphyxiation, cave in, radiation, lighting, housekeeping		20		29	12	3/5	61	Must s
A5 Electrical hazards (severity factors, protection) and biological hazards		12	12			2/5	24	Good t
A9 Fire management in a high-rise building; PASS	8			12		2/5	20	Good t
A3 Numericals: hazard matrix rating, fractional dead time and hazard rate of a protective system			15	8		2/5	23	Good to

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION B: OCCUPATIONAL HEALTH, CHEMICALS, POLLUTION AND ENVIRONMENTAL MA
ASKED PER PAPER

Topic	2024	2023	2022	2021	2020	Years	Marks	
B8 MSDS (material safety data sheet): purpose, 16 headings, key sections, responsibilities of supplier, employer, worker	12	20	12	19	12	5/5	75	Must st
B11 Greenhouse effect and global warming, acid rain, smog: causes, effects, control	8	16	10	20	12	5/5	66	Must st
B6 Hazardous chemicals: categories and health effects; hazardous vs toxic; routes of entry; local/systemic and acute/chronic effects	17	6	21	8	12	5/5	64	Must st
B1 Occupational health, OSH Act and OSHA, occupational health service (functions, how to provide)	12	12	6	9	20	5/5	59	Must st
B2 Occupational diseases: definitions, vs normal disease; lung disease, blood disorders, urticaria, pigmentary changes, sore eye, stress	9	8	15	11	10	5/5	53	Must st
B3 Occupational ergonomics: definition, importance, five examples of good ergonomics	8	9	7	5	5	5/5	34	Must st
B12 Sustainable development: definition, environmental/economic/socio-political sustainability, pollution prevention and clean technology, challenges	12	8	10	8		4/5	38	Must s
B4 Occupational exposure, TLV and PEL, TLV-TWA, TLV-STEL, TLV-C	6	7	14	10		4/5	37	Must s
B15 ISO 14001: purpose and major components of an environmental management system	12	6	7	5		4/5	30	Must s

B10 Local, regional and global environmental impacts and their causes (thermal power plants, petroleum industry)	6	9	6	5		4/5	26	Must s
B13 IEE and EIA: why needed, differences, EIA process (flow chart), red category, EMP	10	16	12	22	8	5/5	68	Must stu
B14 Environmental clearance in Bangladesh (Green, Orange-A, Orange-B, Red), Department of Environment (DoE) and its history	15	7	6	10		4/5	38	Must stu
B9 Pollution and pollutants: classification, sources of air and water pollution and their control		8	7	10	5	4/5	30	Mus
B7 Dose-response relationship; lethal, sub-lethal, acute		8		8	6	3/5	22	Must
B5 Numerical: time-weighted average (TWA) exposure and comparison with TLV	13					1/5	13	Good t

Section A: Occupational safety, hazards, fire, risk, PPE, accidents and EPP

questions 1-4 (1-3 in 2020)

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

A14 Causes of accidents: Heinrich's domino theory, 88-10-2 rule, man-man, man-machine and man-environment relationships

Must study · Very likely · 100% · Easy

Asked in 5 of 5 papers · 52 marks in total

STUDY THESE

- "Accidents do not happen, they are caused": Heinrich found 88 % unsafe acts, 10 % unsafe conditions, 2 % unpreventable
- Domino theory: ancestry/social environment → fault of person → unsafe act/condition → accident → injury; remove one domino (usually the unsafe act/condition) and the injury is prevented
- Three relationships: man-man, man-machine, man-environment; if any is out of balance → unsafe act/condition → accident

> `2024` `Q4(a)` `10 marks`
>
> Briefly explain Heinrich's domino theory of accident. Also explain the man to machine relationship of an accident.

> `2023` `Q4(a)` `7 marks`
>
> How does Heinrich's domino theory explain the causation of accidents in the workplace?

> `2022` `Q3(b)` `13 marks`
>
> What are the major causes of workplace accidents? Explain the accident that occurs due to imbalance of the man and machine relationship. Suggest the remedies.

> `2021` `Q2(c)` `10 marks`
>
> Explain the three basic relationships between man, machine and environment. What are the remedial measures to ensure safety?

> `2020` `Q3(d)` `12 marks`
>
> What are the major causes of workplace accidents? Explain the man and machine relationship.

Full answer: ESE 4125 Study Pack, topic A14.

A1 Basic definitions: hazard, risk, accident, occupational safety, health, safety management; the three parts of OSHEM

Must study · Very likely · 100% · Easy

Asked in 5 of 5 papers · 49 marks in total

STUDY THESE

- Hazard = a condition or act that CAN cause harm (to people, property, environment) but has not yet done so; risk = probability of the harm happening

together with how bad it would be; accident = the unplanned event that actually causes the harm
- OSHEM has three parts: occupational safety management (stop injury and damage now), occupational health management (stop slow illness), environment management (stop pollution of air, water, soil)
- Safety management = systematic process of identifying, assessing and controlling risks so that accidents, injuries and illnesses are prevented

> 2024 Q1(a) 12 marks
>
> Define and explain: (i) Hazard, (ii) Occupational risk, (iii) Occupational accident, (iv) Safety management.

> 2023 Q1(a) 12 marks
>
> Define and explain: (i) Hazard, (ii) Occupational accident, (iii) Occupational safety, (iv) Safety management.

> 2023 Q1(b) 7 marks
>
> How do occupational safety, health and environmental management practices intersect to ensure comprehensive industrial workplace safety? Explain.

> 2022 Q5(d) 7 marks
>
> Explain the necessity of study of the subject "Safety and Environmental Aspects of Energy Projects".

> 2021 Q1(a) 5 marks
>
> What are the components of OSHEM? Which areas do they address?

> 2020 Q1(a) 6 marks
>
> Define: hazard, occupational accident and occupational safety.

Full answer: ESE 4125 Study Pack, topic A1.

A15 Accident investigation and reporting sheet; accident statistics AFR, ASR, FAR (with numerical); HAZAN, HAZOP; safety audit

Must study · Very likely · 93% · Easy

Asked in 4 of 5 papers · 71 marks in total

STUDY THESE

- Investigation answers who, what, when, where, why and how to prevent; the reporting sheet records person, time, place, injuries, losses, events, causes, recommendations
- AFR = injuries × 10^6 / man-hours worked; ASR = days (hours) lost × 1000 / man-hours; FAR = deaths × 10^8 / man-hours
- Man-hours = workers × weeks × hours per week, minus absence

> `2024` `Q4(b)` `10 marks`
>
> Prepare an accident reporting sheet for accidental investigation.

> `2024` `Q4(c)` `5 marks`
>
> Define: (i) accident frequency rate, (ii) accident severity rate, (iii) fatal accident rate, (iv) HAZAN, (v) Triple Bottom Line (TBL).

> `2023` `Q4(b)` `7 marks`
>
> What are the purposes of accident report analysis? Explain AFR, ASR and FAR with their significance.

> `2022` `Q3(c)` `12 marks`
>
> Briefly explain accident investigation and reporting. Prepare a simple accident reporting sheet.

> `2022` `Q4(c)` `12 marks`
>
> Define and explain: (i) AFR, (ii) ASR, (iii) FAR, (iv) HAZAN, (v) HAZOP, (vi) EPP.

> `2021` `Q4(b)` `9 marks`
>
> What are the different types of safety audits? Explain the important aspects to be kept during execution of a safety audit.

> `2021` `Q4(c)` `6 marks`
>
> Define: (i) accident frequency rate, (ii) accident severity rate, (iii) fatal accident rate.

> `2021` `Q4(d)` `10 marks`
>
> An undertaking with 600 workers, working 50 weeks a year with 40 working hours a week, had 60 accidents; the workers were absent during 5 % of the aggregate working time. Find the accident frequency rate.

Full answer: ESE 4125 Study Pack, topic A15.

Sources of hazard, categories of hazard by working situation,

Very likely · 93% Easy

A2 consequences, control steps, psychosocial hazards

Must study

Asked in 4 of 5 papers · 63 marks in total

STUDY THESE

- Five categories by working situation: physical, chemical, biological, psychosocial, human-factor failure
- Sources: dangerous substances, fire/explosion, machinery, electricity, radiation, overhead work, noise/vibration, heat/humidity, dust/fumes. Consequences: human, economic, environmental
- Hazard control in six steps: identify, make inventory, rank, assess probability, assess rank, eliminate/reduce/control

> 2024 Q1(b) 12 marks
>
> Explain the various categories of hazards according to working situation. What are the major sources of hazards and their effects?

> 2023 Q1(d) 6 marks
>
> What are the psychosocial hazards and how do they affect workplace safety?

> 2022 Q1(a) 10 marks
>
> Define hazard. What are the typical sources of hazards and consequences of them?

> 2022 Q1(b) 15 marks
>
> Explain different categories of hazards according to the working situation. (numerical part in A3)

> 2022 Q1(c) 10 marks
>
> Discuss the measures to control the hazards.

> 2021 Q1(b) 10 marks
>
> Define hazard. What are the typical sources of hazard? What are the consequences of hazard?

Full answer: ESE 4125 Study Pack, topic A2.

A10 Hazard identification methods: checklist, what-if, PHA,

Must study · Very likely · 87% · Medium

HAZOP, FMEA, plant drawings; limitations of hazard evaluation

Asked in 4 of 5 papers · 55 marks in total

STUDY THESE

- Hazard identification answers "what can go wrong?" before it happens; methods: checklist, what-if, PHA, HAZOP, FMEA, fault tree, event tree
- PHA = early, rough list of hazards at the concept stage (hazard → cause → effect → category → measures). HAZOP = detailed, line-by-line team study using guide words (NO, MORE, LESS, REVERSE ...) on a P&ID
- Inputs: plot plan, isometric, BFD, PFD, P&ID. Limitations: never complete, depends on team experience, subjective, results hard to reproduce

> `2024` `Q2(c)` `10 marks`
> How do plot plans, isometric drawings, BFDs, PFDs and P&IDs provide the essential system information needed for effective hazard identification?

> `2023` `Q3(d)` `5 marks`
> Differentiate between Preliminary Hazard Analysis (PHA) and HAZOP analysis.

> `2022` `Q4(a)` `12 marks`
> Make a list of various kinds of hazard identification methodologies. (anatomy part in A11)

> `2022` `Q4(b)` `11 marks`
> A design feeds hydrogen sulfide (H_2S) from a cylinder to a processing unit. The analyst knows only that H_2S is toxic and flammable and identifies its potential release as hazardous. Do a preliminary hazard analysis (PHA).

> `2020` `Q1(b)` `12 marks`
> What are the methods of hazard identification? Describe the HAZOP procedure.

> `2020` `Q3(a)` `5 marks`
> What are the limitations of a hazard evaluation study?

Full answer: ESE 4125 Study Pack, topic A10.

A13 Personal protective equipment (PPE): types, how each protects, respiratory protective equipment

Must study · Likely · 73% · Easy

Asked in 4 of 5 papers · 38 marks in total

STUDY THESE

- PPE = equipment worn by the worker; it is the LAST control, used when the hazard cannot be removed
- Head (hard hat), eye/face (glasses, goggles, face shield, welding shield), ear (plugs, muffs), respiratory (air-purifying or air-supplying), hand (gloves), foot (safety shoes), body (coverall, apron, suit), fall (harness)
- Respirators: air-purifying (dust mask, cartridge/gas mask, PAPR) clean the air; atmosphere-supplying (SCBA, supplied-air) give clean air - needed in oxygen-deficient or IDLH air

> `2024` `Q3(a)` `10 marks`
>
> What are the various PPEs used in industry and how do they protect humans against various hazards associated with industry?

> `2022` `Q3(a)` `10 marks`
>
> How is PPE used in general industry to protect against various hazards associated with humans?

> `2021` `Q4(a)` `10 marks`
>
> What is meant by PPE? Briefly write about respiratory protective equipment.

> `2020` `Q3(b)` `8 marks`
>
> What are the common PPEs used in industry? Mention their advantages.

Full answer: ESE 4125 Study Pack, topic A13.

A7 Classes of fire and fire-fighting materials (water, foam, dry powder, CO_2, wet chemical); fire-fighting procedure

Must study · Likely · 67% · Easy

Asked in 4 of 5 papers · 50 marks in total

STUDY THESE

- Class A ordinary solids, B flammable liquids, C flammable gases, D metals, E electrical, F/K cooking oil
- Water for A only (never on oil, metal, live electrics); foam A+B; dry powder almost all incl. D and electrical; CO_2 electrical + B; wet chemical F

- Procedure: RACE (Rescue, Alarm, Contain, Extinguish) and PASS for the extinguisher (Pull, Aim, Squeeze, Sweep)

> 2023 Q2(b) 13 marks
>
> Define fire and flame. Briefly explain the fire-fighting procedure and fire-fighting materials for each class of fire.

> 2022 Q2(b) 13 marks
>
> Briefly explain the fire classes based on the types of fuel involved, and give an overview of commonly used fire-fighting materials for each class.

> 2021 Q3(b) 10 marks
>
> What are the fire-fighting materials used in industry? Can water be used to extinguish all types of fire? Explain.

> 2020 Q2(c) 14 marks
>
> Discuss the applications of fire-fighting materials for different classes of fire.

Full answer: ESE 4125 Study Pack, topic A7.

A16 Emergency preparedness plan (EPP), Triple Bottom Line (TBL), safety committee

Must study · Likely · 67% · Easy

Asked in 3 of 5 papers · 36 marks in total

STUDY THESE

- EPP = written plan of what everyone does in an emergency; 11 steps from listing possible disasters to evacuation; 4 disaster levels
- TBL = judge a business by 3 Ps: People (social), Planet (environment), Profit (economy) - John Elkington, 1997
- Safety committee = management + worker representatives; chairman (manager), member secretary (safety officer), factory doctor, personnel officer, foremen, worker (union) representatives

> 2024 Q4(d) 10 marks
>
> Briefly explain the steps of preparing an Emergency Preparedness Plan (EPP).

> 2023 Q4(c) 7 marks
>
> What is the Triple Bottom Line (TBL) framework? Who are the typical members of a safety committee and what roles do they play?

> 2023 Q4(d) 14 marks
>
> What are the steps for an emergency preparedness plan? Briefly explain them.

> 2020 Q3(c) 5 marks
>
> Classify the EPP (emergency preparedness plan).

Full answer: ESE 4125 Study Pack, topic A16.

A11 Anatomy of a process incident; event tree vs fault tree

Must study · Likely · 67% · Medium

Asked in 3 of 5 papers · 31 marks in total

STUDY THESE

- A process drifts from normal → abnormal → emergency; each mode has its plant state, operating goal, critical systems and support activities
- Normal: optimise, keep normal. Abnormal: return to normal, establish control. Emergency: bring to safe state, minimise losses
- Fault tree works backward from a top event (deduction, AND/OR gates); event tree works forward from an initiating event (induction, success/failure branches) - they are logical counterparts

> 2024 Q3(b) 13 marks
>
> Briefly explain the anatomy of a process incident with their modes, states, operational goals and critical systems.

> 2022 Q4(a) 12 marks
>
> Describe the anatomy of a process incident mentioning their different modes, states and goals.

> 2021 Q3(c) 6 marks
>
> "Event tree analysis is the logical counterpart of the fault tree" - explain.

Full answer: ESE 4125 Study Pack, topic A11.

A8 BLEVE (boiling liquid expanding vapour explosion): mechanism, sequence of events, effects, prevention

Must study · Likely · 67% · Medium

Asked in 3 of 5 papers · 28 marks in total

STUDY THESE

- A tank of liquefied gas (LPG) is heated by fire; pressure rises, wall above the liquid overheats and weakens, tank splits
- Sudden pressure drop makes the superheated liquid flash-boil and expand ~250 times → blast + fireball + flying fragments
- Prevent: cool the tank with water spray, insulation, relief valve, depressurise, slope/drain spills away, keep distance

> 2024 Q2(b) 10 marks
>
> Explain the mechanism of BLEVE with the sequence of events leading to the BLEVE.

> 2022 Q2(c) 10 marks
>
> What is BLEVE? "All combustible materials do not create explosion" - explain.

> 2021 Q2(d) 8 marks
>
> Define BLEVE. Why is prevention of BLEVE necessary?

Full answer: ESE 4125 Study Pack, topic A8.

A12 Risk management, hierarchy of controls (and where PPE fits), bow-tie diagram

Must study · Likely · 60% · Medium

Asked in 2 of 5 papers · 49 marks in total

STUDY THESE

- Risk management = identify hazards → estimate consequence and likelihood → evaluate risk → reduce it → monitor
- Hierarchy (most → least effective): elimination, substitution, engineering control, administrative control, PPE (last line of defence)
- Bow-tie: threats (left) → prevention barriers → TOP EVENT (loss of control) → mitigation barriers → consequences (right); degradation factors and controls on barriers

> 2024 Q2(d) 7 marks
>
> Briefly explain the correct order of the risk control hierarchy from most effective to least effective.

> 2024 Q3(c) 12 marks

> Describe the structure of a bow-tie diagram with the identification of key elements.

> 2023 Q3(a) 10 marks
> Define risk management. Briefly explain the key steps in the hazard identification process.

> 2023 Q3(b) 10 marks
> What is a bow-tie diagram and how is it used in risk management? Explain.

> 2023 Q3(c) 10 marks
> What is the hierarchy of mitigation controls for managing workplace hazards and how do different types of PPE fit into it?

Full answer: ESE 4125 Study Pack, topic A12.

A6 Fire basics: fire triangle and tetrahedron, factors contributing to fire, methods of extinguishment, types of explosion

Must study · Possible · 53% · Easy

Asked in 3 of 5 papers · 34 marks in total

STUDY THESE

- Fire = rapid oxidation that gives heat and light; needs fuel + oxygen + heat (triangle) + chain reaction (tetrahedron)
- Extinguish by removing one side: starving (remove fuel), smothering (cut oxygen), cooling/quenching (remove heat), inhibition (break chain reaction)
- Explosion types: dust explosion, deflagration (flame slower than sound), detonation (faster than sound), vapour cloud explosion (UVCE/CVCE), BLEVE

> 2024 Q1(c) 11 marks
> What are the different types of explosion? Explain the different methods of fire extinguishment.

> 2021 Q3(a) 7 marks
> "All combustible materials are not fuel" - explain.

> 2020 Q2(a) 6 marks
> What are the methods of fire extinguishment? Define each.

> 2020 Q2(b) 10 marks

> Explain the fire triangle. What factors contribute to a fire incident?

Full answer: ESE 4125 Study Pack, topic A6.

A4 Physical and thermal hazards: heat cramps, heat stroke, frostbite, asphyxiation, cave in, radiation, lighting, housekeeping

Must study · Possible · 47% · Easy

Asked in 3 of 5 papers · 61 marks in total

STUDY THESE

- Heat: heat cramps (salt loss → muscle spasm), heat exhaustion (fatigue, clammy skin), heat stroke (body temperature control fails → emergency). Cold: hypothermia (whole body), frostbite (tissue freezes locally)
- Asphyxiation = oxygen cannot reach body tissues (simple: N_2, CO_2, methane dilute O_2; chemical: CO, H_2S block blood). Cave in = buried in loose grain, sand, sawdust
- Radiation: ionizing (X-ray, isotopes → cancer), limit 5 rem/year whole body; UV from welding → eye. Bad housekeeping → slips, trips, cuts, fire

> `2023 Q1(c) 10 marks`
>
> Briefly explain the thermal hazards that workers face in extreme temperature conditions. Also discuss their effects.

> `2023 Q2(c) 10 marks`
>
> Briefly explain: (i) heat cramps, (ii) cave in, (iii) asphyxiation, (iv) frostbite, (v) heat stroke.

> `2021 Q1(d) 12 marks`
>
> Write down the types of hazards imposed by radiation. Are there any limits for radiation exposure? What are the preventive measures?

> `2021 Q2(a) 10 marks`
>
> Short notes: (i) emergency lighting, (ii) heat cramps, (iii) cave in, (iv) asphyxiation.

> `2021 Q2(b) 7 marks`
>
> What do you understand by improper housekeeping? What types of accidents can occur due to improper housekeeping?

> `2020 Q1(c) 12 marks`
>
> Explain: frostbite, heat stroke, asphyxiation and ionizing radiation.

Full answer: ESE 4125 Study Pack, topic A4.

A5 Electrical hazards (severity factors, protection) and biological hazards

Good to have · Possible · 47% · Easy

Asked in 2 of 5 papers · 24 marks in total

STUDY THESE

- Severity of electric shock depends on voltage, body resistance (current = V/R), path through the body and duration; 10-20 mA = cannot let go, 50-100 mA = heart fibrillation
- Protection: insulation, guarding, grounding (earthing), circuit protection (fuse, circuit breaker, GFCI), safe work practice (lockout/tagout), PPE, training
- Biological hazards = bacteria, viruses, fungi, parasites; high-risk jobs: hospital, laboratory, sewage, waste, farming/animals

> 2023 Q2(a) 12 marks
>
> Briefly explain the main methods of protection against electrical hazards in industry.

> 2022 Q2(a) 12 marks
>
> What are the key factors that determine the severity and consequences of electrical injuries? What are the risks posed by biological hazards in different occupational settings?

Full answer: ESE 4125 Study Pack, topic A5.

A9 Fire management in a high-rise building; PASS

Good to have · Possible · 47% · Easy

Asked in 2 of 5 papers · 20 marks in total

STUDY THESE

- Three parts: fire prevention, fire fighting, evacuation
- Fire fighting: detectors and alarm, sprinklers, hydrants, water reserve, fire pumps (main, standby, jockey), firemen lift, extinguishers, trained team, drills
- Evacuation: fire-proof stair with fire doors, never use lift, stay low under smoke, assembly point, head count

> 2024 Q2(a) 8 marks
>
> Properly explain the fire management procedure in a high rise building.

> 2021 Q3(d) 12 marks
>
> What are meant by PASS and fire triangle? Explain the fire management in high-rise buildings.

Full answer: ESE 4125 Study Pack, topic A9.

A3 Numericals: hazard matrix rating, fractional dead time and hazard rate of a protective system

Good to have · Possible · 33% · Medium

Asked in 2 of 5 papers · 23 marks in total

STUDY THESE

- Hazard (risk) rating = severity rating (20, 15, 10, 3) × frequency rating (10, 8, 6, 4, 2): 100-200 unacceptable, 60-90 undesirable, 20-50 conditionally acceptable, 6-18 acceptable
- Fractional dead time FDT = failure rate × test interval / 2 (the fraction of time the safety device is secretly broken)
- Hazard rate = demand rate × FDT (how often a demand arrives while the device is dead)

> 2022 Q1(b) 15 marks
>
> A safety valve has a failure rate of once every two years (0.5/year), the test interval is 1 week and the system pressure is expected to rise once a year. Calculate the fractional dead time and hazard rate.

> 2021 Q1(c) 8 marks
>
> How could you assess the impacts of hazard with the help of a hazard matrix?

Full answer: ESE 4125 Study Pack, topic A3.

Section B: Occupational health, chemicals, pollution and environmental management

questions 5-8 (4-6 in 2020)

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

B8 MSDS (material safety data sheet): purpose, 16 headings, key sections, responsibilities of supplier, employer, worker

Must study · Very likely · 100% · Easy

Asked in 5 of 5 papers · 75 marks in total

STUDY THESE

- MSDS = document from the supplier telling the hazards of a chemical and how to use, store, handle it and what to do in an emergency
- 16 headings; key ones: 1 identification, 2 hazards, 4 first aid, 5 fire-fighting, 7 handling & storage, 8 exposure control/PPE, 11 toxicology, 15 regulatory
- Supplier prepares and gives it; employer keeps it updated (≤ 3 years old), accessible and trains workers; worker reads it and follows it

> 2024 Q6(c) 12 marks
> Discuss the purpose of MSDS. Also discuss six crucial contents to prepare an MSDS.

> 2023 Q6(c) 6 marks
> What is the primary purpose of a Material Safety Data Sheet (MSDS) and why is it important for work safety?

> 2023 Q6(d) 14 marks
> Discuss these terms to prepare an MSDS: (i) identification of the substance, (ii) hazards identification, (iii) first aid measures, (iv) handling and storage, (v) toxicological information, (vi) regulatory information.

> 2022 Q7(a) 12 marks
> Briefly discuss MSDS. Explain the responsibilities of the (i) supplier, (ii) employer, (iii) worker.

> 2021 Q6(a) 8 marks
> What is meant by hazardous chemicals? What are the responsibilities of a supplier and an employer related to MSDS?

> 2021 Q6(c) 11 marks

What is meant by MSDS? Discuss: (i) identification of substance and company, (ii) hazards identification, (iii) first aid measures, (iv) handling and storage, (v) regulatory information.

2020 Q5(c) 12 marks

What are the crucial contents of an MSDS?

Full answer: ESE 4125 Study Pack, topic B8.

B11 Greenhouse effect and global warming, acid rain, smog: causes, effects, control

Must study · Very likely · 100% · Easy

Asked in 5 of 5 papers · 66 marks in total

STUDY THESE

- Greenhouse effect: sunlight warms the Earth, the Earth gives off infrared heat, greenhouse gases (CO_2, CH_4, N_2O, CFCs, water vapour) trap part of it → warmer air
- Acid rain: SO_2 and NO_x + water → H_2SO_4, HNO_3; rain pH below 5.6 (down to ~2.4); kills fish, forests, damages buildings
- Smog = smoke + fog: London (sulphurous, from coal) and Los Angeles (photochemical, NO_x + hydrocarbons from cars + sunlight → ozone)

2024 Q7(a) 8 marks

What are the major causes and environmental effects of (i) acid rain, (ii) SMOG?

2023 Q7(d) 10 marks

What are the major greenhouse gases responsible for global warming? Explain the mechanism of the greenhouse effect.

2023 Q8(a) 6 marks

Short notes on (i) acid rain, (ii) SMOG.

2022 Q7(d) 10 marks

How is global warming caused by greenhouse gases? What are the measures to prevent (i) acid rain, (ii) SMOG?

2021 Q7(c) 10 marks

What is the greenhouse effect? How does CO_2 in air increase global warming?

2021 Q7(d) 10 marks

Write the causes and effects of (i) acid rain, (ii) SMOG.

2020 Q6(c) 12 marks

Explain the greenhouse effect, smog and acid rain.

Full answer: ESE 4125 Study Pack, topic B11.

B6 Hazardous chemicals: categories and health effects; hazardous vs toxic; routes of entry; local/systemic and acute/chronic effects

Must study · Very likely · 100% · Easy

Asked in 5 of 5 papers · 64 marks in total

STUDY THESE

- Hazardous chemical = any chemical that can damage health OR cause physical danger (fire, explosion, corrosion); toxic chemical = one that poisons the body - a subset
- Routes: inhalation (breathing - most common), ingestion (swallowing), injection (through a cut/needle), skin or eye absorption
- Local effect at the contact point (acid burn); systemic effect elsewhere (CO → blood). Acute = short high exposure, quick effect; chronic = long low exposure, slow effect (benzene → leukaemia)

2024 Q6(a) 8 marks

Distinguish with examples: (i) systemic and local effects, (ii) acute and chronic effects.

2024 Q6(b) 9 marks

What are the differences between hazardous chemicals and toxic chemicals?
Write the effects of common categories of hazardous chemicals on human health.

2023 Q5(c) 6 marks

Distinguish between inhalation, ingestion and injection.

2022 Q6(c) 9 marks

Write the common categories of hazardous chemicals and explain their effects on health. How do toxic substances affect the surroundings?

2022 Q6(d) 12 marks

Define and explain with examples: (i) inhalation, (ii) ingestion, (iii) systemic effects, (iv) chronic effects.

2021 Q6(d) 8 marks

Write the occupational effects of: (i) toxic, (ii) corrosive, (iii) reactive, (iv) flammable chemicals.

2020 Q5(b) 8 marks

Explain local, systemic, acute and chronic effects.

2020 Q5(d) 4 marks

What are the types of hazardous chemical exposure process?

Full answer: ESE 4125 Study Pack, topic B6.

B1 Occupational health, OSH Act and OSHA, occupational health service (functions, how to provide)

Must study · Very likely · 100% · Easy

Asked in 5 of 5 papers · 59 marks in total

STUDY THESE

- Occupational health = keeping workers physically, mentally and socially well and protecting them from illness caused by work (ILO/WHO)
- OSH Act 1970 (USA) is the law; OSHA is the agency that sets and enforces the standards (PELs). In Bangladesh: Labour Act 2006
- Occupational health service (ILO Convention 161, 1985): preventive service - risk assessment, monitoring, medical examinations, first aid, advice, training, records, rehabilitation

2024 Q5(a) 12 marks

Short notes: (i) occupational health, (ii) occupational safety act, (iii) occupational diseases, (iv) occupational exposure.

2023 Q5(a) 12 marks

Define occupational health and occupational diseases. What are the functions of an occupational health service?

2022 Q5(a) 6 marks

Short notes: (i) occupational health, (ii) Occupational Safety and Health Act.

2021 Q5(a) 9 marks

Short notes: (i) occupational health, (ii) Occupational Safety and Health Act, (iii) Occupational Safety and Health Administration.

> 2020 Q4(a) 10 marks
>
> Define occupational health, occupational diseases, occupational exposure and TLV.

> 2020 Q4(b) 10 marks
>
> How can proper occupational health be provided?

Full answer: ESE 4125 Study Pack, topic B1.

B2 Occupational diseases: definitions, vs normal disease; lung disease, blood disorders, urticaria, pigmentary changes, sore eye, stress

Must study · Very likely · 100% · Easy

Asked in 5 of 5 papers · 53 marks in total

STUDY THESE

- Occupational disease = disease caused by exposure at work to a hazardous material, physical agent, biological organism or ergonomic stress
- Lung: dusts (silica, coal, asbestos), fumes, isocyanates, cotton dust. Blood: benzene (aplasia, leukaemia), lead (anaemia), aniline/nitrites (methaemoglobinaemia), arsine (haemolysis)
- Skin: urticaria (itchy red welts, latex), pigmentary changes (tar, UV). Eye: VDU work, welding, dust, chlorine. Stress: workload, shift, bullying, insecurity

> 2024 Q5(b) 9 marks
>
> What are the major occupational causes of: (i) lung disease, (ii) blood disorders, (iii) pigmentary changes?

> 2023 Q5(b) 8 marks
>
> Explain: (i) urticaria, (ii) pigmentary changes, (iii) blood disorders, (iv) sore eye.

> 2022 Q5(b) 15 marks
>
> Discuss the differences between occupational diseases and normal diseases. Write the major causes of: (i) lung disease, (ii) sore eye, (iii) blood disorders, (iv) urticaria.

> 2021 Q5(b) 11 marks
>
> Provide two definitions of occupational disease. Write the occupational causes of: (i) stress reactions, (ii) lung diseases, (iii) blood disorders.

> 2020 Q4(c) 10 marks
>
> Explain urticaria and pigmentary changes.

Full answer: ESE 4125 Study Pack, topic B2.

B3 Occupational ergonomics: definition, importance, five examples of good ergonomics

Must study · Very likely · 100% · Easy

Asked in 5 of 5 papers · 34 marks in total

STUDY THESE

- Ergonomics (Greek ergon = work, nomos = law) = the science of making the job fit the worker
- Importance: fewer musculoskeletal injuries (back, neck, wrist), less fatigue and error, more productivity and comfort
- Examples: adjustable chair with back support, screen at eye level, tools within easy reach, lifting aids, anti-fatigue mats, good lighting, job rotation and breaks

> 2024 Q5(c) 8 marks
>
> Define occupational ergonomics. Mention five examples of good occupational ergonomics and explain them.

> 2023 Q5(d) 9 marks
>
> Define occupational ergonomics. Mention five examples of good occupational ergonomics and explain them.

> 2022 Q5(c) 7 marks
>
> Define occupational ergonomics. Explain its importance in the workplace.

> 2021 Q5(d) 5 marks
>
> Briefly discuss occupational ergonomics.

> 2020 Q6(a) 5 marks
>
> Why is ergonomics important?

Full answer: ESE 4125 Study Pack, topic B3.

Sustainable development: definition,

B12 environmental/economic/socio-political sustainability, pollution prevention and clean technology, challenges

Must study · Very likely · 93% · Easy

Asked in 4 of 5 papers · 38 marks in total

STUDY THESE

- Sustainable development = development that meets the needs of the present without compromising the ability of future generations to meet their own needs (Brundtland, 1987)
- Three pillars: environmental (protect resources), economic (viable, profitable), socio-political (fair, participatory, stable)
- Challenges: population growth, urbanisation/industrialisation, intensive agriculture, declining resources, altered biogeochemical cycles, social disintegration

> 2024 Q7(b) 12 marks
>
> Define sustainable development. Discuss: (i) environmental sustainability, (ii) economic sustainability, (iii) socio-political sustainability.

> 2023 Q7(c) 8 marks
>
> What is meant by sustainable development? Briefly discuss the challenges of sustainable development.

> 2022 Q8(a) 10 marks
>
> "Sustainable development calls for pollution prevention and clean technology" - explain. How to overcome the challenges of sustainable development?

> 2021 Q8(a) 8 marks
>
> Why does sustainable development call for pollution prevention and clean technology? What are the environmental challenges of sustainable development?

Full answer: ESE 4125 Study Pack, topic B12.

B4 Occupational exposure, TLV and PEL, TLV-TWA, TLV-STEL, TLV-C

Must study · Very likely · 93% · Easy

Asked in 4 of 5 papers · 37 marks in total

STUDY THESE

- Occupational exposure = how much (concentration) and how long (time) a worker is in contact with a hazard
- TLV (ACGIH) = recommended airborne limit almost all workers can face day after day without harm; PEL (OSHA) = legal limit. Same idea, TLV advisory, PEL law
- TLV-TWA = 8 h average (40 h/week); TLV-STEL = 15-minute limit (max 4 times/day, 60 min apart); TLV-C = ceiling, never exceed even for an instant

> 2024 Q5(d) 6 marks
>
> Write down the information that can be obtained from: (i) TLV-TWA, (ii) TLV-STEL, (iii) TLV-C.

> 2023 Q6(b) 7 marks
>
> Why is it necessary to study PEL and TLV? Explain the relation between them.

> 2022 Q6(a) 8 marks
>
> What is meant by occupational exposure? What are the two terms that reflect the level of exposure? Differentiate between them.

> 2022 Q6(b) 6 marks
>
> Write down the information that can be obtained from: (i) TLV-TWA, (ii) TLV-STEL, (iii) TLV-C.

> 2021 Q5(c) 10 marks
>
> Write the differences between TLVs and PELs. Explain the different ways of calculating the TLVs.

Full answer: ESE 4125 Study Pack, topic B4.

B15 ISO 14001: purpose and major components of an environmental management system

Must study · Very likely · 93% · Easy

Asked in 4 of 5 papers · 30 marks in total

STUDY THESE

- ISO 14001 = international standard for an environmental management system (EMS); the only ISO 14000 standard a company can be certified to
- Purpose: identify and control environmental aspects, comply with law, prevent pollution, improve continually, gain trust and save resources
- Built on Plan-Do-Check-Act: policy and leadership, aspects and legal requirements, objectives, programmes, training, operation and emergency,

monitoring and audit, corrective action, management review

> 2024 Q8(b) 12 marks
>
> Mention the purpose of ISO 14001. What are the major components of ISO 14001?

> 2023 Q8(e) 6 marks
>
> Briefly discuss ISO 14001.

> 2022 Q8(d) 7 marks
>
> Mention the purposes of ISO 14001. What are the major components of ISO 14001?

> 2021 Q8(d) 5 marks
>
> Briefly discuss ISO 14001.

Full answer: ESE 4125 Study Pack, topic B15.

B10 Local, regional and global environmental impacts and their causes (thermal power plants, petroleum industry)

Must study · Very likely · 93% · Easy

Asked in 4 of 5 papers · 26 marks in total

STUDY THESE

- Local: urban air pollution (SO_2, CO, NO_x, particulates, ground ozone, heavy metals), indoor air, polluted rivers/groundwater/soil
- Regional: acid rain, pollution of seas and cross-border waters
- Global: greenhouse effect/climate change, persistent pollutants and radioactive waste, loss of biodiversity, ocean acidification, desertification

> 2024 Q6(d) 6 marks
>
> Explain the following terms of environmental impact and their main causes: (i) local, (ii) regional, (iii) global.

> 2023 Q7(b) 9 marks
>
> Explain local, regional and global environmental impacts and mention their main causes.

> 2022 Q7(c) 6 marks
>
> Explain the following terms of environmental impact: (i) local, (ii) regional, (iii) global.

2021 Q7(b) 5 marks

What are the local and regional environmental impacts posed by thermal power plants and petroleum industries?

Full answer: ESE 4125 Study Pack, topic B10.

B13 IEE and EIA: why needed, differences, EIA process (flow chart), red category, EMP

Must study · Very likely · 100% · Medium

Asked in 5 of 5 papers · 68 marks in total

STUDY THESE

- IEE (initial environmental examination) = first, quick screening at pre-feasibility; EIA (environmental impact assessment) = full, detailed study at feasibility with quantified impacts and mitigation
- Bangladesh (ECA 1995, ECR 1997): Red category projects need IEE for site clearance and EIA for environmental clearance from DoE
- EIA process: screening → IEE → scoping/ToR → full EIA → review → decision → monitoring and audit. EMP = the plan to carry out mitigation and monitoring during construction and operation

2024 Q7(c) 10 marks

Why are IEE and EIA studies required for a project? Explain the EIA process.

2023 Q8(b) 10 marks

When are IEE and EIA studies required for a project? Explain the EIA process in detail.

2023 Q8(c) 6 marks

What is an Environmental Management Plan (EMP)? At which stage of a new project is EMP necessary and why?

2022 Q8(b) 12 marks

Why are IEE and EIA studies required for a project? What type of assessment is necessary for a red-category project? Explain with a flow chart.

2021 Q8(b) 12 marks

What are the differences between EIA and IEE? With a flow chart, explain the EIA process.

2021 Q8(c) 10 marks

Why is an Environmental Management Plan essential during project implementation? (clearance part in B14)

2020 Q6(d) 8 marks

What are the purposes of IEE and EIA?

Full answer: ESE 4125 Study Pack, topic B13.

B14 Environmental clearance in Bangladesh (Green, Orange-A, Orange-B, Red), Department of Environment (DoE) and its history

Must study · Very likely · 93% · Medium

Asked in 4 of 5 papers · 38 marks in total

STUDY THESE

- Environmental clearance certificate (ECC) = DoE permission to set up or run an industry/project (ECA 1995 section 12, ECR 1997 rule 7)
- Steps: apply to DoE (Form-3, fee, documents, NOC) → site clearance → apply for ECC → ECC → renew yearly (Green every 3 years)
- Red: feasibility report + IEE with ToR for EIA → site clearance → submit EIA + ETP design → approval → ECC (within 60 days)

2024 Q7(d) 5 marks

Write down the brief history of the Department of Environment (DoE) of Bangladesh.

2024 Q8(a) 10 marks

Explain the general procedure to get environmental clearance for a red category new project according to the regulation Act in Bangladesh.

2023 Q8(d) 7 marks

Discuss the general process of gaining environmental clearance of a new project.

2022 Q8(c) 6 marks

Write the procedure to get environmental clearance for a new project according to the regulation Act in Bangladesh.

2021 Q8(c) 10 marks

Discuss the general process of gaining environmental clearance of a new project.

Full answer: ESE 4125 Study Pack, topic B14.

B9 Pollution and pollutants: classification, sources of air and water pollution and their control

Must study · Likely · 67% · Easy

Asked in 4 of 5 papers · 30 marks in total

STUDY THESE

- By nature of disposal: persistent (slowly degradable: pesticides), biodegradable (sewage, garbage), non-biodegradable (lead, mercury, plastic, nuclear waste)
- By existence: quantitative (natural gases like CO_2 that become pollutants above a limit) and qualitative (man-made: pesticides)
- Air sources: fossil fuel burning (power plants, vehicles, industry) → CO, CO_2, SO_2, NO_x, particulates. Water sources: industrial effluent, sewage, farm runoff, oil spills

> 2023 Q7(a) 8 marks
>
> What are the major sources of water pollution? How to control them?

> 2022 Q7(b) 7 marks
>
> Classify the materials that create pollution, with definition and examples.

> 2021 Q7(a) 10 marks
>
> Classify and explain the pollutants. Discuss the major sources of air pollution.

> 2020 Q6(b) 5 marks
>
> Classify pollutants by the nature of disposal.

Full answer: ESE 4125 Study Pack, topic B9.

B7 Dose-response relationship; lethal, sub-lethal, acute

Must study · Possible · 47% · Easy

Asked in 3 of 5 papers · 22 marks in total

STUDY THESE

- Dose = amount a person receives; response = effect on the body; bigger dose → bigger effect
- Curve: below threshold no effect → reversible effects → permanent damage → death

- Lethal = dose that kills; sub-lethal = below lethal, harms growth/reproduction/behaviour; acute = effect within a short time after one exposure

> 2023 Q6(a) 8 marks
>
> What is meant by hazardous chemicals? Explain the dose-response relationship. Show (i) lethal, (ii) sub-lethal, (iii) acute in it.

> 2021 Q6(b) 8 marks
>
> Explain the hazardous chemical dose-response relationship. Point out (i) lethal, (ii) sub-lethal and (iii) acute.

> 2020 Q5(a) 6 marks
>
> Explain the dose-response relationship.

Full answer: ESE 4125 Study Pack, topic B7.

B5 Numerical: time-weighted average (TWA) exposure and comparison with TLV

Good to have · Possible · 33% · Easy

Asked in 1 of 5 papers · 13 marks in total

STUDY THESE

- TWA = $\Sigma(C_i\ t_i)$ / 8 h, where C_i is the concentration during period t_i
- Periods with no exposure (lunch away from the area) count as C = 0
- If TWA ≤ TLV-TWA the worker is within the limit

> 2024 Q8(c) 13 marks
>
> TLV-TWA of a substance is 200 ppm. Shift 8:00 AM - 5:00 PM (9 h) with 1 h lunch (12:00-1:00 PM) with no exposure. Hourly concentrations (ppm) at 8:00, 9:00 ... 16:00: 137, 130, 129, 157, 136, 138, 160, 140, 147. (i) Calculate the TWA, (ii) has the worker exceeded the TLV? (iii) Comment on the effect of exposure time and concentration on TLV-TWA.

Full answer: ESE 4125 Study Pack, topic B5.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
3R	reduce, reuse, recycle
ACGIH	American Conference of Governmental Industrial Hygienists
AFR	accident frequency rate
AIT	auto-ignition temperature
ANSI	American National Standards Institute
APR	air-purifying respirator
ASR	accident severity rate
BFD	block flow diagram
BLEVE	boiling liquid expanding vapour explosion
CAS	Chemical Abstracts Service (registry number of a chemical)
CB	circuit breaker
CCl_4	carbon tetrachloride
CEO	chief executive officer
CFC	chlorofluorocarbon
CNG	compressed natural gas
CPSC	Consumer Product Safety Commission (USA)
CSR	corporate social responsibility
CVCE	confined vapour cloud explosion
dB	decibel (unit of sound level)
dBA	A-weighted decibel (sound level as the human ear hears it)
DDT	dichloro-diphenyl-trichloroethane (an insecticide)
DEPC	Department of Environmental Pollution Control
DIFE	Department of Inspection for Factories and Establishments (Bangladesh)
DoE	Department of Environment (Bangladesh)
DOT	Department of Transportation (USA)
ECA	Environment Conservation Act (Bangladesh, 1995)
ECC	Environmental Clearance Certificate
ECR	Environment Conservation Rules (Bangladesh, 1997; new rules 2023)

EIA	environmental impact assessment
EMP	environmental management plan
EMS	environmental management system
EPA	Environmental Protection Agency (USA)
EPP	emergency preparedness plan
EPS	earnings per share
ESP	electrostatic precipitator
ETA	event tree analysis
ETP	effluent treatment plant
FAR	fatal accident rate
FDT	fractional dead time
FMEA	failure modes and effects analysis
FTA	fault tree analysis
GFCI	ground-fault circuit interrupter
GHG	greenhouse gas
GHS	Globally Harmonised System of classification and labelling of chemicals
HAZAN	hazard analysis
HAZOP	hazard and operability study
HFC	hydrofluorocarbon
HI	hazard identification
HIRA	hazard identification and risk assessment
HR	human resources
ICC	International Chamber of Commerce
ICRP	International Commission on Radiological Protection
IDLH	immediately dangerous to life or health
IEE	initial environmental examination
ILO	International Labour Organization
IR	infrared
ISO	International Organization for Standardization

LC50	lethal concentration for 50 % of a test population
LD50	lethal dose for 50 % of a test population
LEL	lower explosive limit
LNG	liquefied natural gas
LOTO	lockout/tagout
LPG	liquefied petroleum gas
mA	milliampere = 1/1000 ampere
MIC	methyl isocyanate
MoEF	Ministry of Environment and Forest
MoEFCC	Ministry of Environment, Forest and Climate Change
MSDS	material safety data sheet
mSv	millisievert (radiation dose unit)
NDT	non-destructive testing
NEP	National Environment Policy
NFPA	National Fire Protection Association (USA)
NIOSH	National Institute for Occupational Safety and Health (USA)
NO_x	nitrogen oxides
NOC	no objection certificate
NoC	no objection certificate
NRR	noise reduction rating
NSC	National Safety Council (USA)
OEL	occupational exposure limit
OHS	occupational health and safety
OHSAS	Occupational Health and Safety Assessment Series
OSH	occupational safety and health
OSHA	Occupational Safety and Health Administration (US government agency)
OSHEM	Occupational Safety, Health and Environment Management
P&ID	piping and instrumentation diagram
PAN	peroxyacetyl nitrate

PAPR	powered air-purifying respirator
PASS	Pull, Aim, Squeeze, Sweep (using an extinguisher)
PCB	polychlorinated biphenyl
PDCA	Plan-Do-Check-Act
PEL	permissible exposure limit
PFD	process flow diagram
PHA	preliminary hazard analysis
PM	particulate matter ($PM_{2.5}$ = particles below 2.5 micrometres)
PPE	personal protective equipment
ppm	parts per million
QRA	quantitative risk assessment
RACE	Rescue, Alarm, Contain, Extinguish/Evacuate
REL	recommended exposure limit
rem	roentgen equivalent man (unit of radiation dose; 1 rem = 10 mSv)
RFP	request for proposal
SAR	supplied-air respirator
SCBA	self-contained breathing apparatus
SDS	safety data sheet
SO_x	sulphur oxides
SPM	suspended particulate matter
STEL	short-term exposure limit
TBL	triple bottom line
TDI	toluene diisocyanate
TEDE	total effective dose equivalent
TLV	threshold limit value
ToR	terms of reference
TWA	time-weighted average
UEL	upper explosive limit
UV	ultraviolet

UVCE	unconfined vapour cloud explosion
VCE	vapour cloud explosion
VDU	visual display unit (computer screen)
VOC	volatile organic compound
WHO	World Health Organization

Built from the scanned ESE 4125 papers (2020, 2021, 2022, 2023, 2024). Question text is cleaned up from the scans.